WSe₂ | top3-only repair | absolute meV, per-angle Γ constraints
Calibration: 2.13–5.09°. All metrics below use converged paths, not the coarser fitting basis.

Frozen / repaired candidate	Global coefficients	Mean top3 RMS	Worst top3 RMS	Max Γ top2/3 error	Worst Γ-patch RMS
Previous stronger-gap34 fit	12	1.573	4.324	5.563	3.069
Linear r²: Γ cap 2 meV	12	1.584	2.159	2.000	1.374
Linear r²: Γ cap 1 meV	12	1.636	2.198	1.000	1.160
Quadratic r² diagnostic	17	1.218	2.160	2.002	1.374

Primary candidate retains the original five A+B r² curves (12 global coefficients).
Seven physical Hamiltonian terms. One mass and one λ shared by every angle.
No β, no gap34/P4 target, no band-identity or positive-gap veto.

Γ has improved; not all strict 2meV path goals are met. Models at 6–9° and small-angle extrapolations remain unvalidated/unreliable.

What changed and what did not

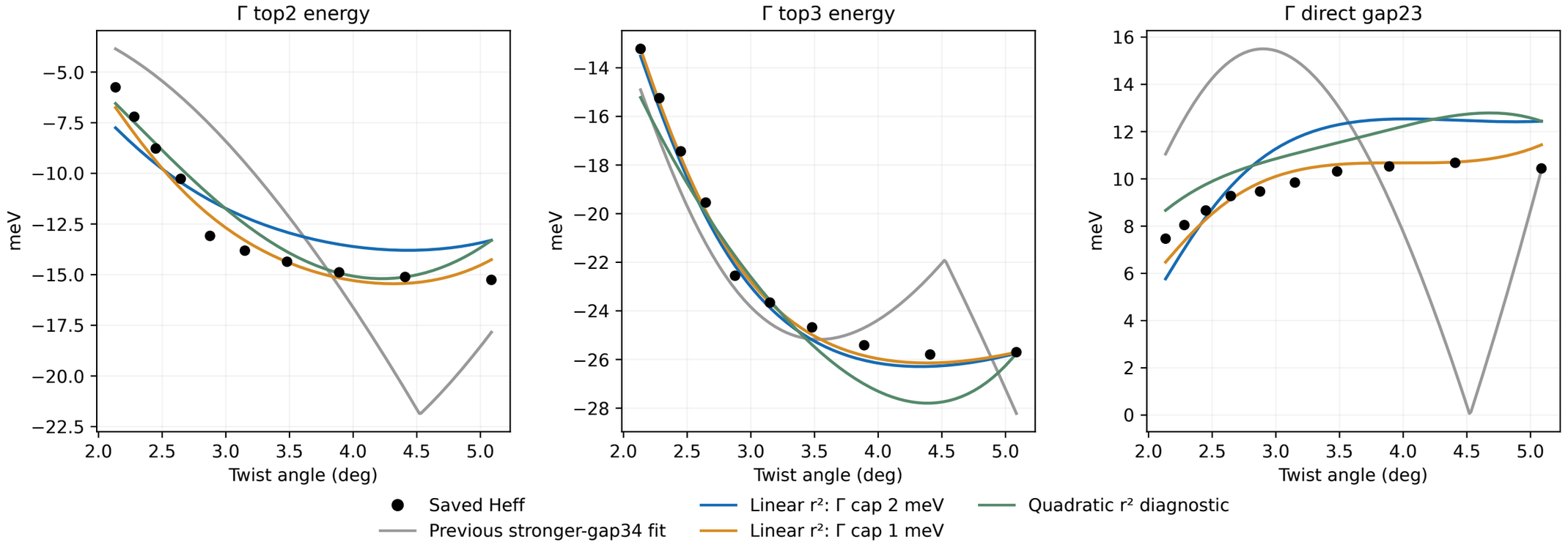
Hamiltonian unchanged: quadratic kinetic + V1/phase, w1, V2, w2, real momentum tunneling λ.
Primary angle functions unchanged: Vcos, Vsin, w1, V2, w2 = A + B r^2.
Only top1–3 are fitted. Band4–6 curves are supplied as visual context.

Optimization: minimize the WORST retained-angle mean top3 spectral error.
Each retained angle has explicit Γ-top2/3 and Γ-gap23 error constraints.
The Γ-neighborhood top2/3 RMS is constrained separately. All energy scales are absolute meV.
One Γ-top1 energy zero per angle; no band-dependent or valley-dependent shifts.
No Chern-number objective, band-parity guard or positive gap34 acceptance veto.

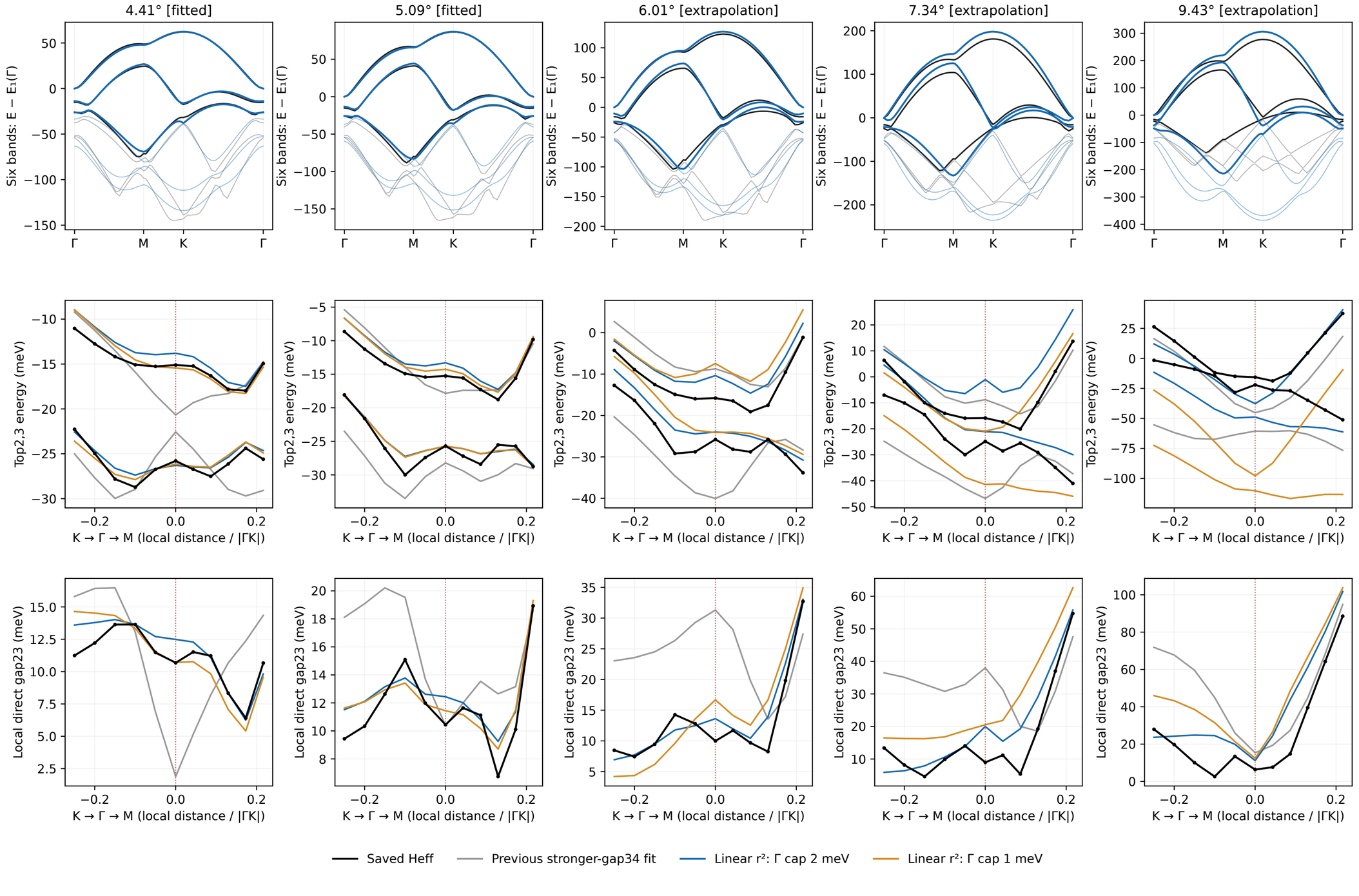
Primary: Γ-energy/gap error <=2meV, training Γ-patch RMS <=2.5meV.
Stricter diagnostic: Γ error <=1meV, training Γ-patch RMS <=2meV.
Worst-angle errors quoted in plots include all 60 unique saved path points.
Training constraints use the original training k subset; unused k points were historically inspected.

Endpoint deletion: electronic data of the deleted angle enter only AFTER fitting.
Initializers use the same retained-angle references or fixed generic constants.
This is retrospective CV, not an unseen-data benchmark. Failed starts remain documented.

All 13 reference angles are shown. 6.01/7.34/9.43 are extrapolations of near-window models.
Independent five-potentials-per-angle and all-angle fits were also attempted; large-angle failures remain.
Below 2.13° there is no own-Heff validation. Deletion spreads are sensitivity, not confidence intervals.
No new Chern or full-BZ extrema calculation is represented by this energy-fit report.

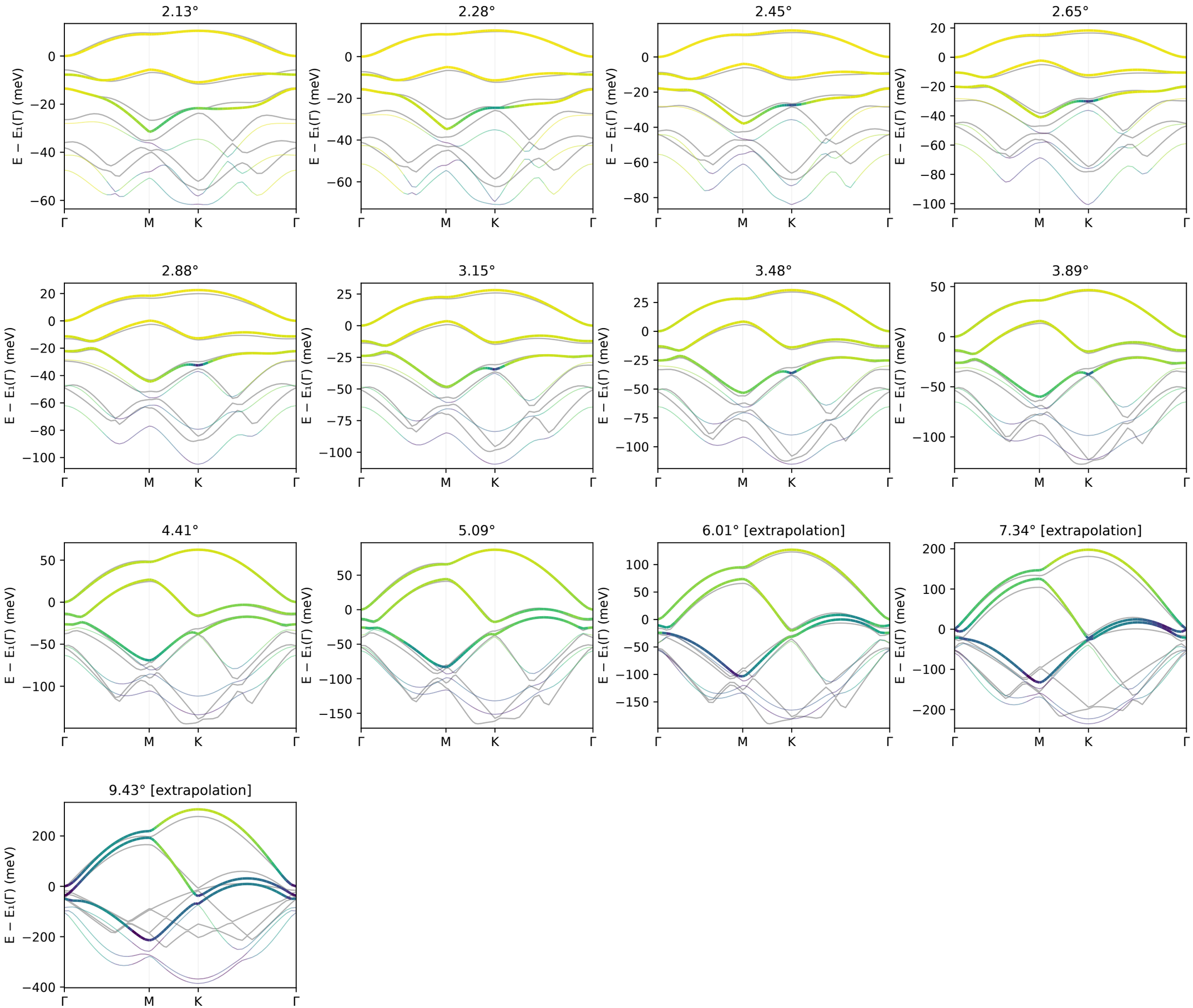

WSe₂ | repairing Γ top2,3 across the fitted interval
241-angle Γ-only scans. Reference dots are calculated Heff values, not a continuous first-principles curve.
Γ top2 energy
Γ top3 energy
Γ direct gap23
meV
Twist angle (deg)
Saved Heff
Previous stronger-gap34 fit
Linear r²: Γ cap 2 meV
Linear r²: Γ cap 1 meV
Quadratic r² diagnostic
A Γ-only gap scan is not a full-BZ gap or Chern calculation. Band exchanges remain allowed.

WSe₂ | Γ-region before/after | extrapolation failures remain visible
Top row shows six-band context for the repair candidate. Lower rows isolate top2,3 and gap23.



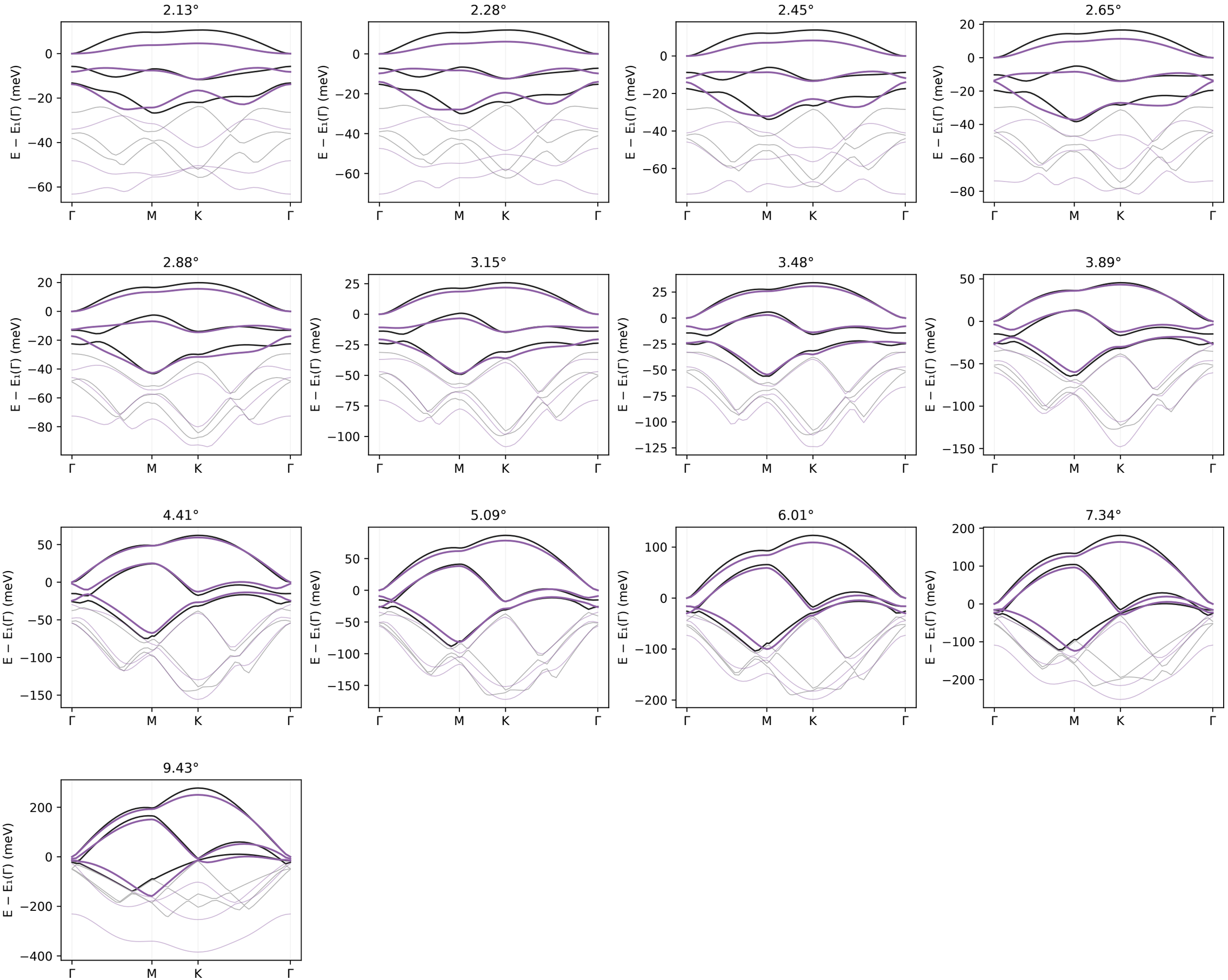
All complete datasets use one Γ-top1 alignment. No band-by-band shifts or band-identity vetoes.

WSe₂ | Linear r²: Γ cap 2 meV | all 13 reference angles
Six K-valley bands. Top3 thick; bands4–6 are thin diagnostic context. Grey curves: saved Heff.



6.01–9.43° remain inaccurate. Numerical convergence does not make the extrapolated spectrum physically reliable.

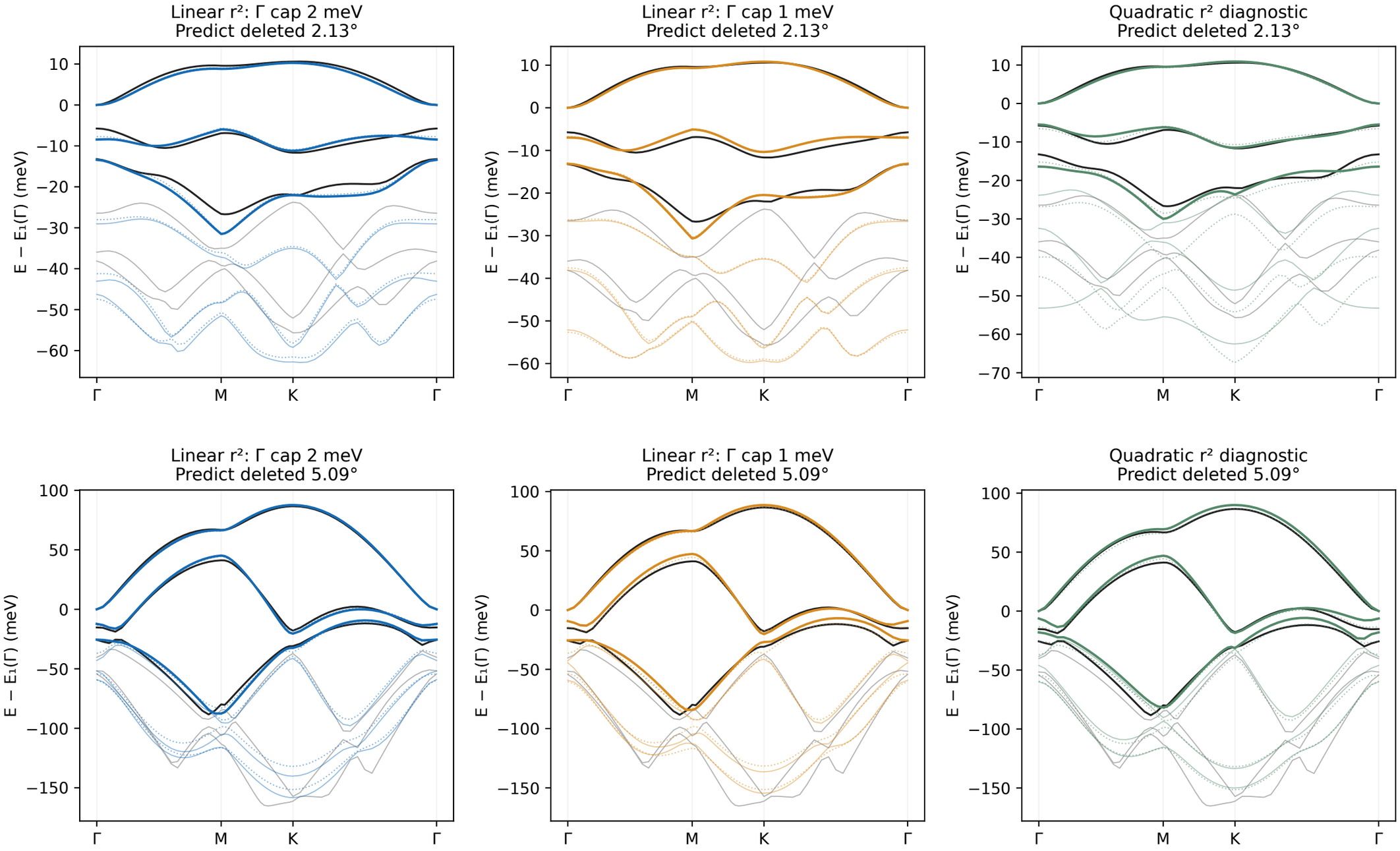
WSe₂ | All-angle fit (fails accuracy goals) | all 13 reference angles
Six K-valley bands. Top3 thick; bands4–6 are thin diagnostic context. Grey curves: saved Heff.



6.01–9.43° remain inaccurate. Numerical convergence does not make the extrapolated spectrum physically reliable.

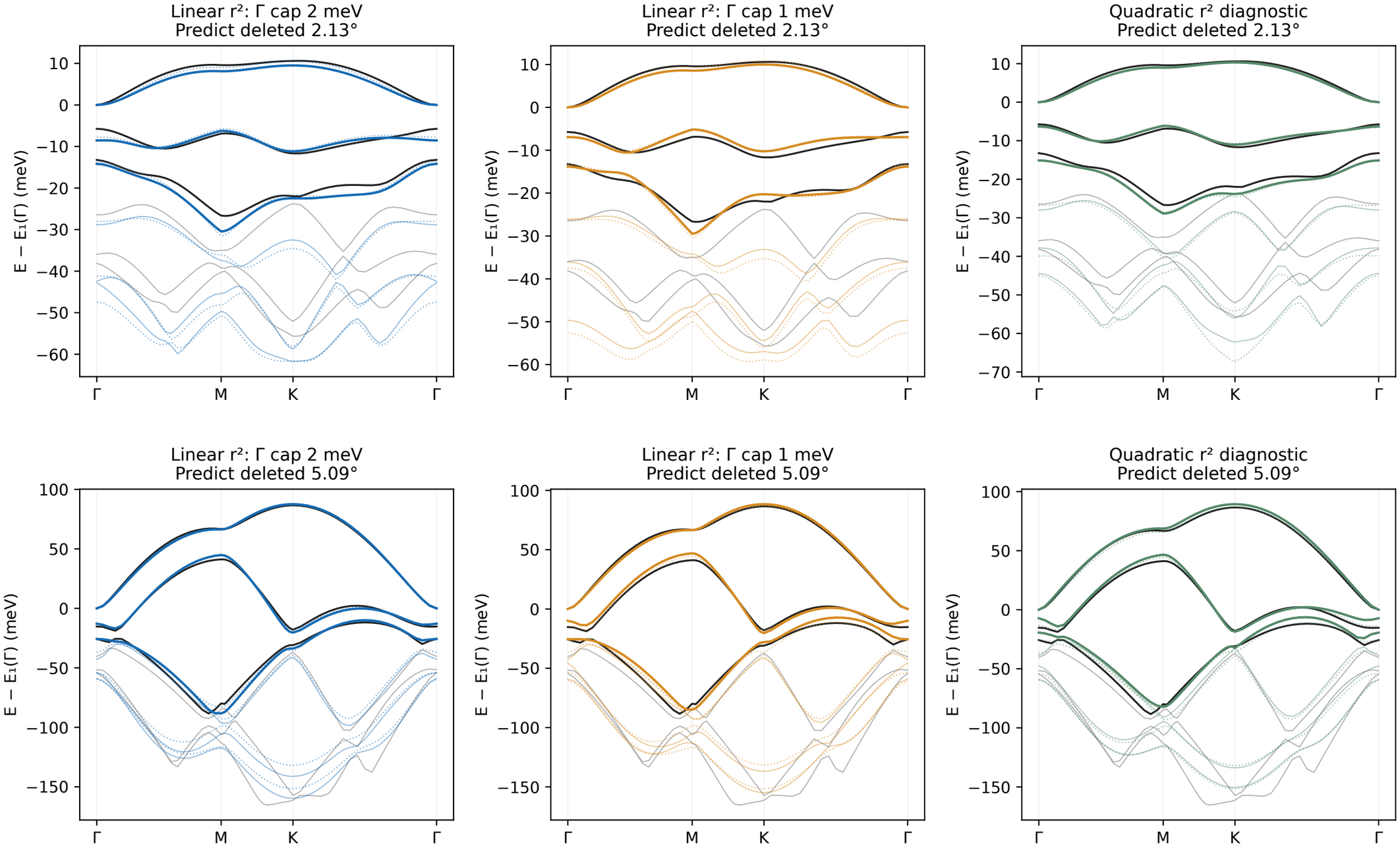
WSe₂ | true endpoint deletion refits

Black: Heff. Solid: deleted-angle prediction. Dotted: full-fit curve. Six K-valley bands; top3 thick.



Each deletion fit and initializer use retained electronic data only. This is retrospective CV, not a blind external benchmark.

WSe₂ | true endpoint deletion refits | delete both endpoints
Black: Heff. Solid: deleted-angle prediction. Dotted: full-fit curve. Six K-valley bands; top3 thick.

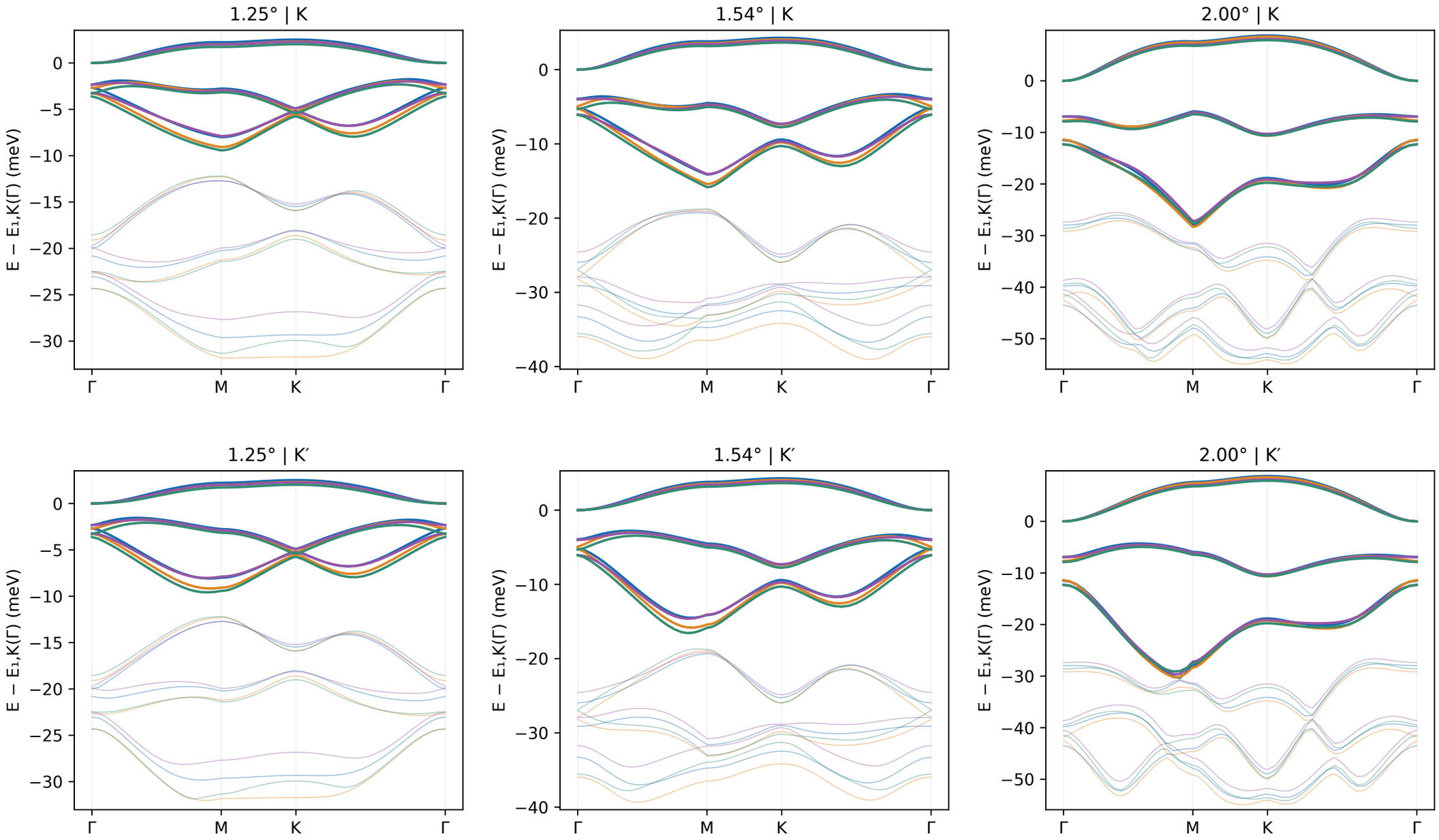


Each deletion fit and initializer use retained electronic data only. This is retrospective CV, not a blind external benchmark.

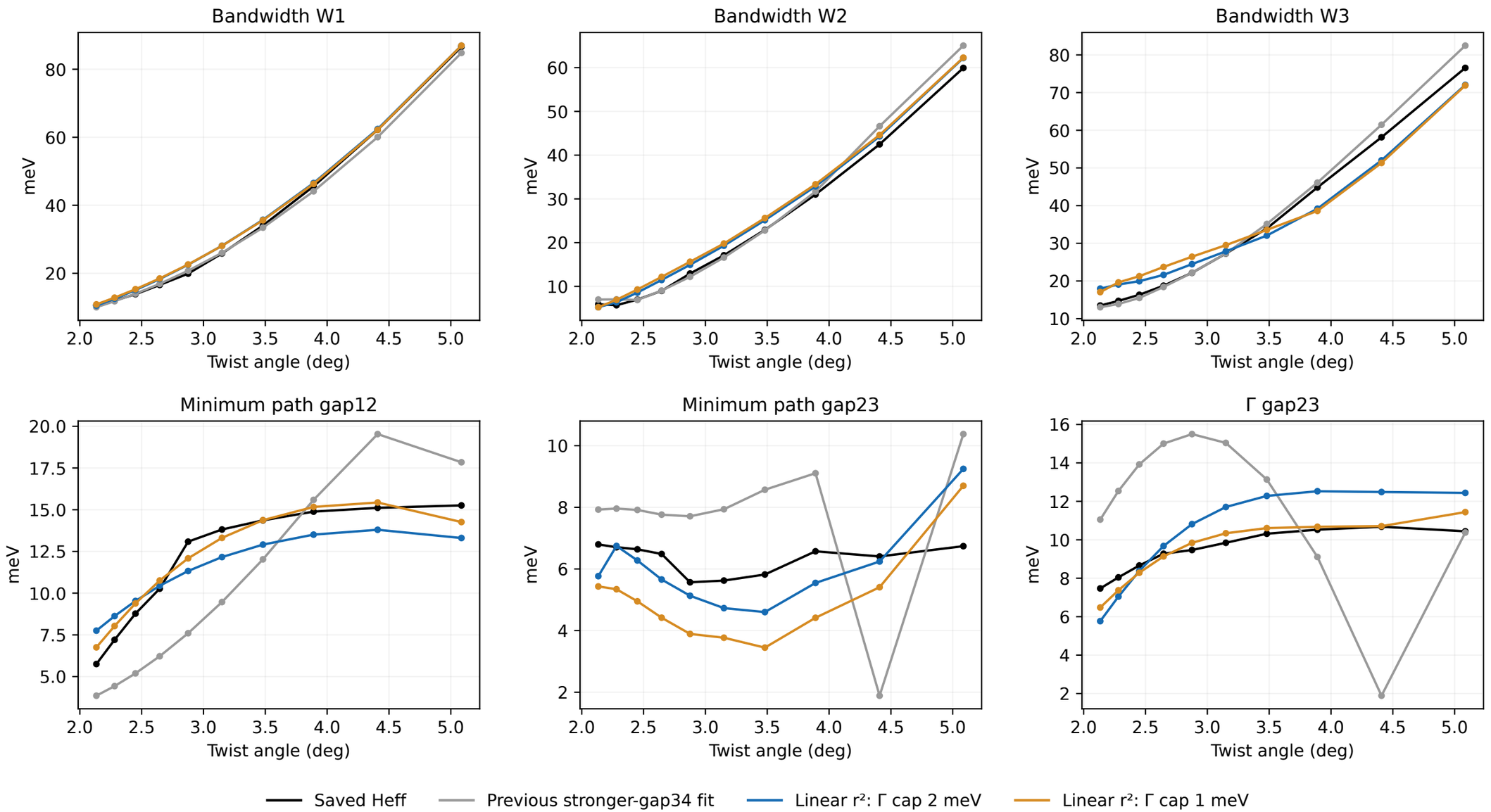
WSe₂ | repaired parameter functions | same seven physical terms
Primary/strict: five A+B r² curves, r=sin(θ/2)/sin(1.5°). Shared mass and λ are constant across angles.

Grey region: unvalidated small-angle extrapolation. Quadratic angular controls do not add Hamiltonian terms, but worsen endpoint robustness here.

WSe₂ | small-angle prediction sensitivity | six bands per valley
Same simple primary model, refitted after deleting one or both endpoints. Top3 thick; lower bands are context.



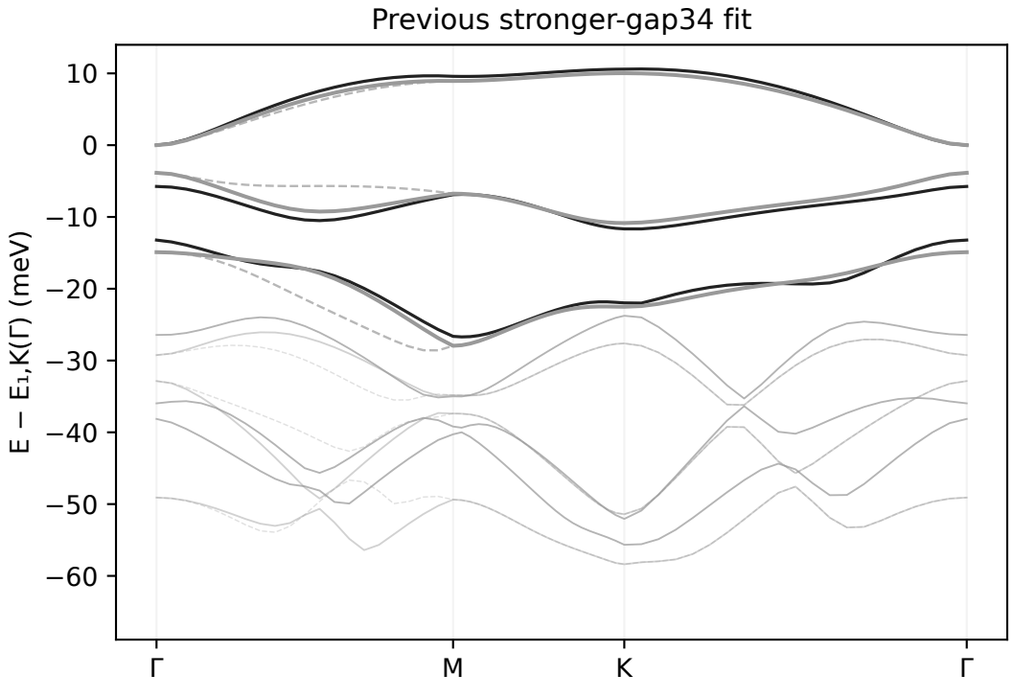
No independent own-Heff reference below 2.13°. Curve spreads are sensitivity, not errors or confidence intervals.

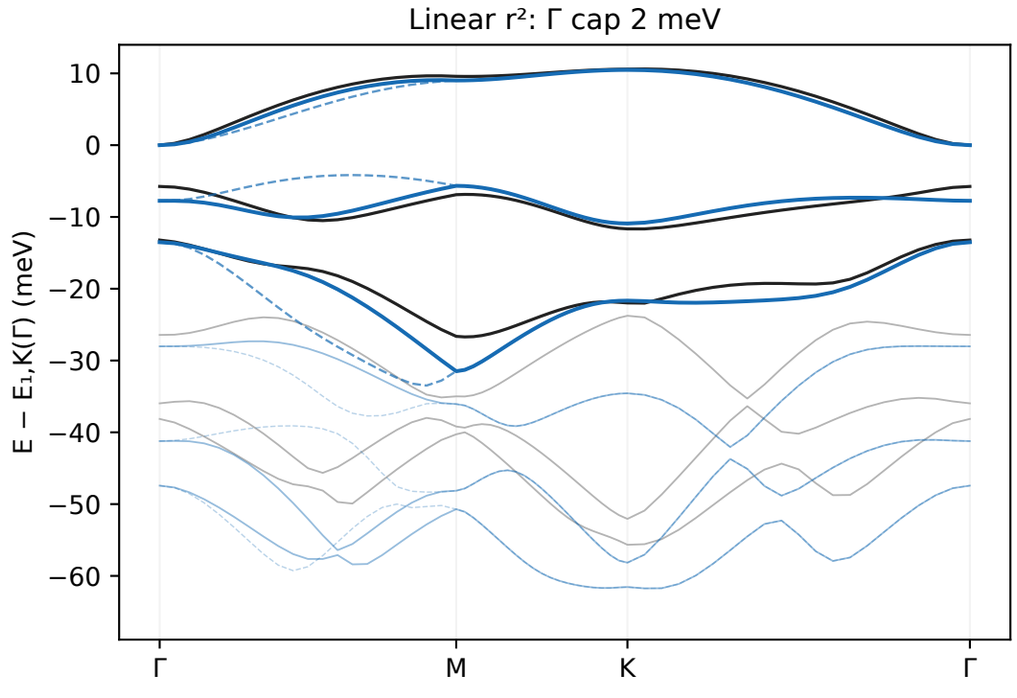

WSe₂ | top3 bandwidths and gaps on the saved path
Reference and all models use the SAME path scope. These are not full-BZ extrema; no Chern inference.
Bandwidth W1
Bandwidth W2
Bandwidth W3
Minimum path gap12
Minimum path gap23
Γ gap23
meV
Twist angle (deg)
Saved Heff
Previous stronger-gap34 fit
Linear r²: Γ cap 2 meV
Linear r²: Γ cap 1 meV

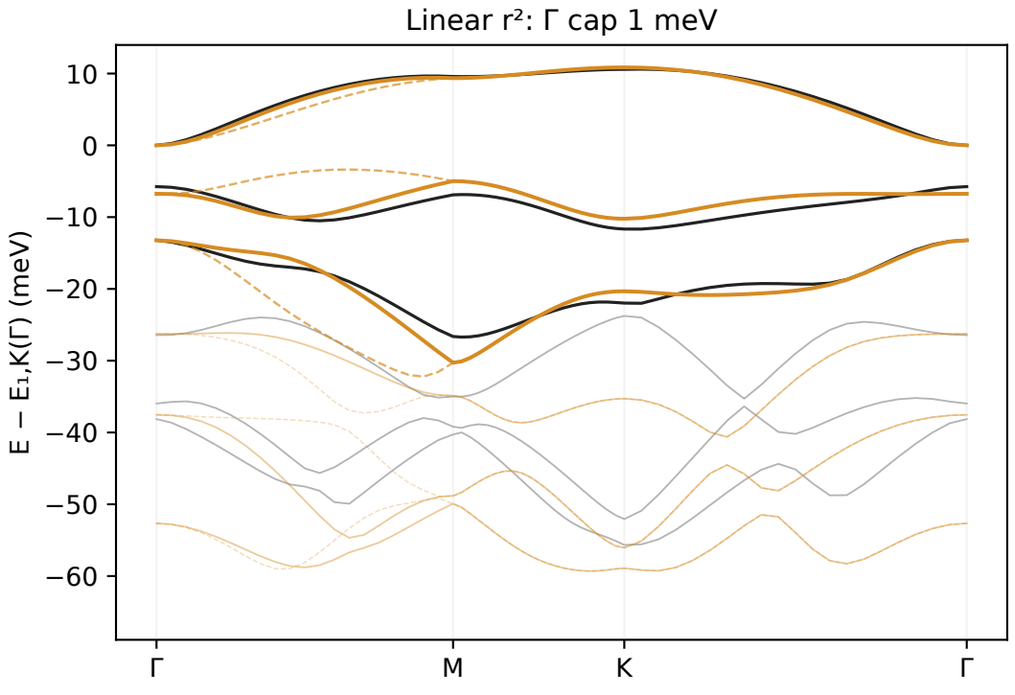
WSe₂ | genuine deleted-angle prediction errors | meV
Full-data fits are not used to initialize deletion fits. Quadratic functions can fit known points better but extrapolate worse.

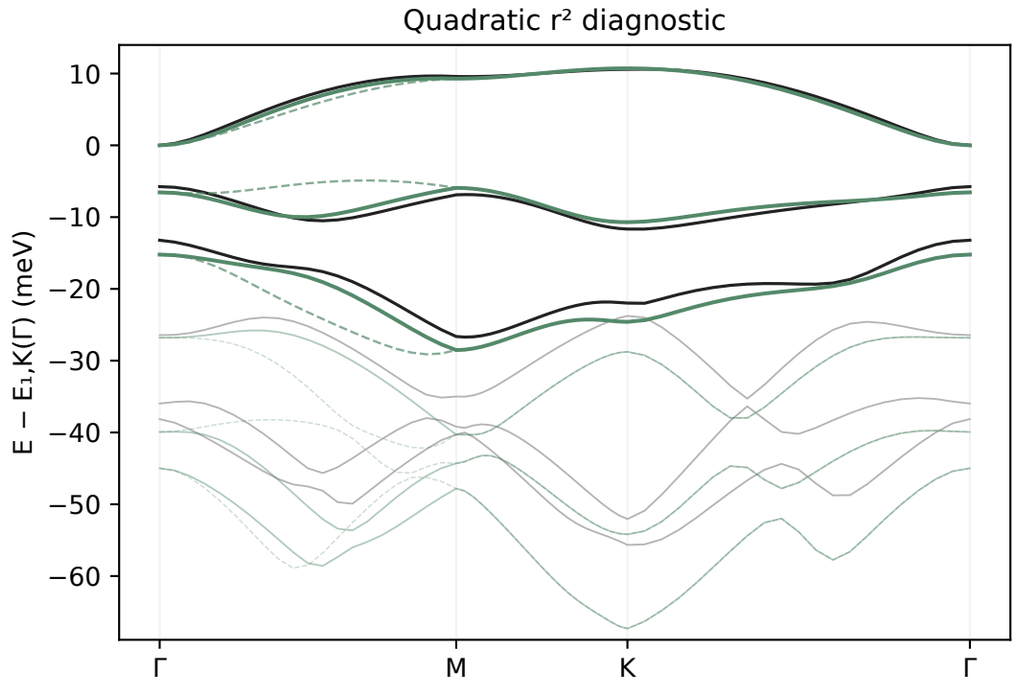
Candidate	Deleted angles	Target θ	Top3 RMS	Γ E2 error	Γ E3 error	Γ-patch RMS
robust	omit_low	2.13	1.51	-2.69	-0.18	1.39
robust	omit_high	5.09	2.66	+2.99	+0.18	1.90
robust	omit_both	2.13	1.46	-2.79	-0.94	1.63
robust	omit_both	5.09	2.63	+2.39	-0.02	1.56
flexible	omit_low	2.13	1.19	+0.29	-3.18	1.76
flexible	omit_high	5.09	4.24	+8.93	+7.60	6.06
flexible	omit_both	2.13	0.99	-0.55	-1.89	1.10
flexible	omit_both	5.09	3.77	+8.02	+6.33	5.31
strict	omit_low	2.13	1.15	-1.23	+0.09	0.67
strict	omit_high	5.09	3.56	+5.98	+0.07	4.12
strict	omit_both	2.13	1.07	-1.19	-0.58	0.68
strict	omit_both	5.09	3.33	+5.33	+0.11	3.67
old	omit_low	2.13	0.79	+1.23	-1.04	0.90
old	omit_high	5.09	4.43	+2.30	-8.59	4.82
old	omit_both	2.13	0.79	+1.50	-1.19	1.05
old	omit_both	5.09	5.03	-6.28	+1.46	2.62

WSe₂ | 2.13393° | top3 repair with six-band context
All own models use pure quadratic kinetic energy and common mass/λ. Solid K; dashed K′; reference K only.



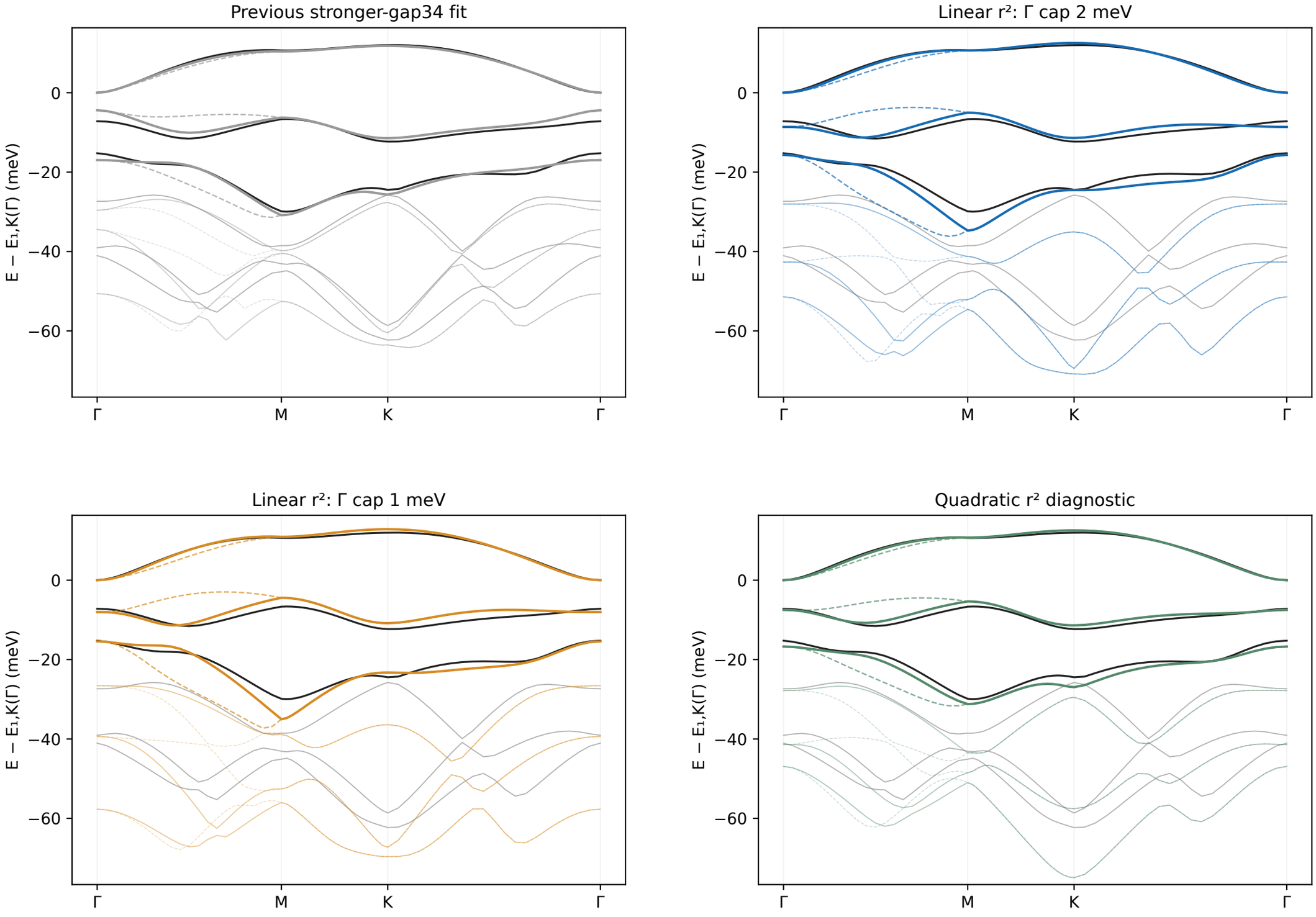






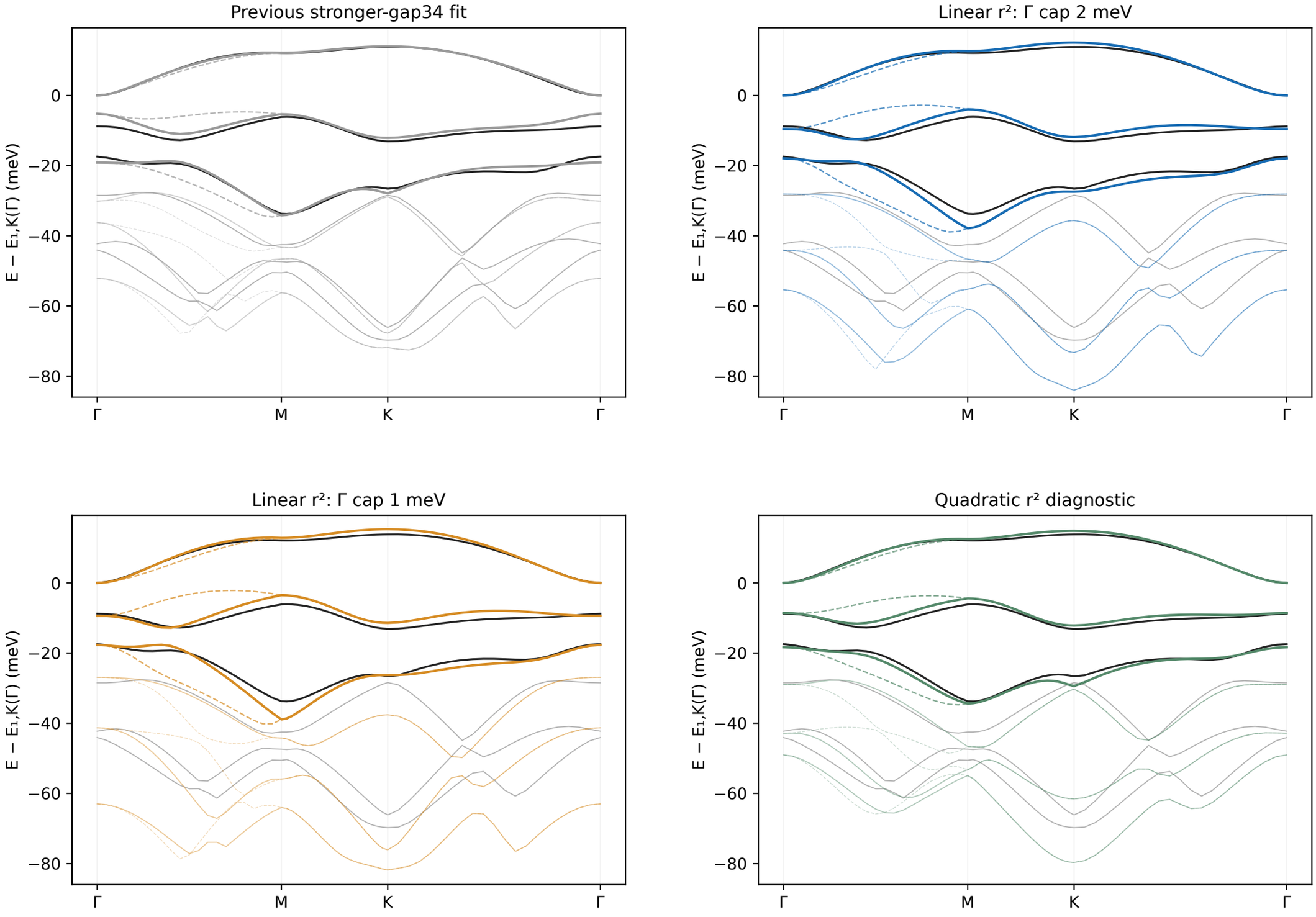
Top3 are thick and fitted. Bands4–6 are thin context. Opposite-valley Heff is not available in these snapshots.

WSe₂ | 2.28106° | top3 repair with six-band context
All own models use pure quadratic kinetic energy and common mass/λ. Solid K; dashed K′; reference K only.



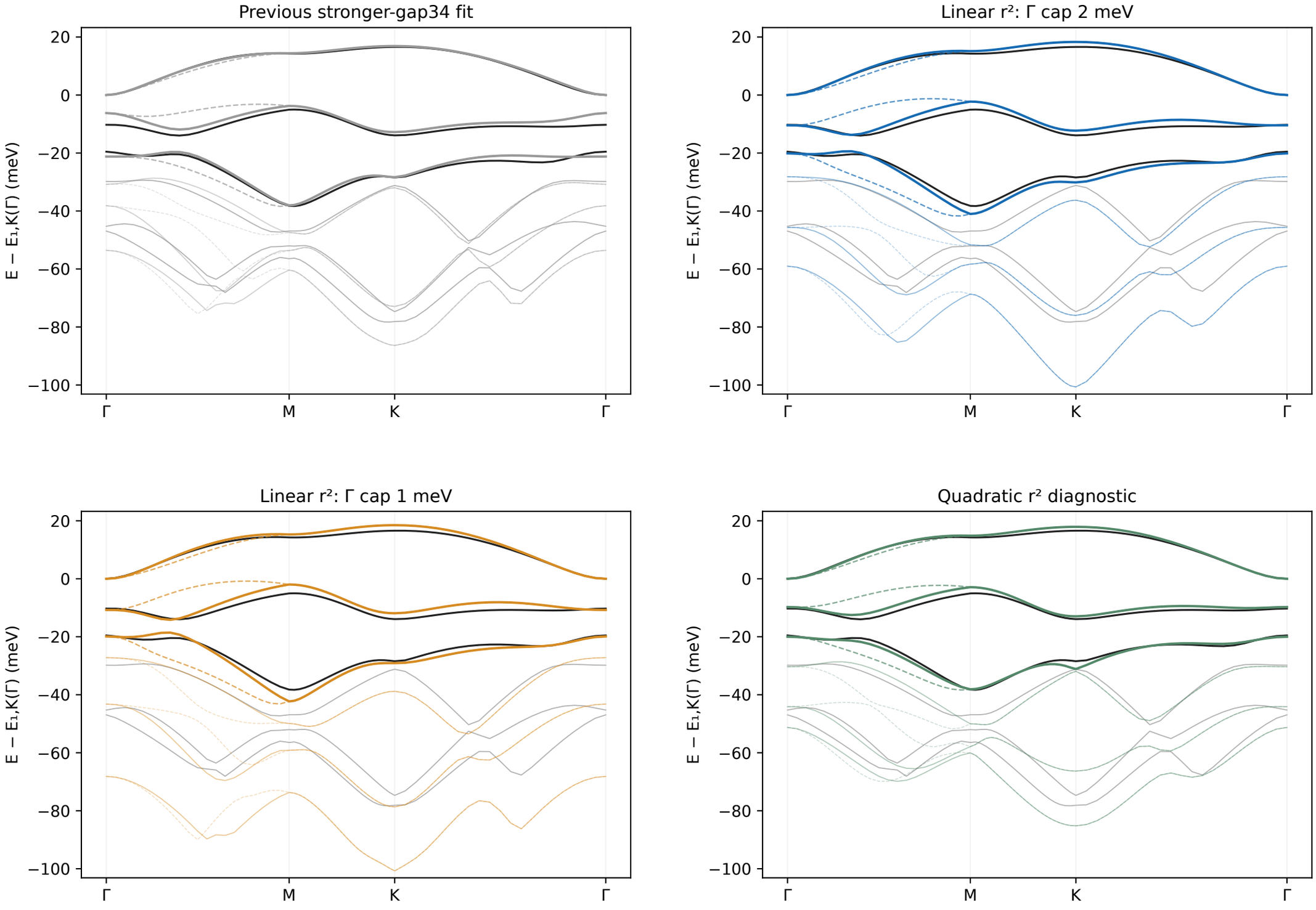
Top3 are thick and fitted. Bands4–6 are thin context. Opposite-valley Heff is not available in these snapshots.

WSe₂ | 2.44998° | top3 repair with six-band context
All own models use pure quadratic kinetic energy and common mass/λ. Solid K; dashed K′; reference K only.



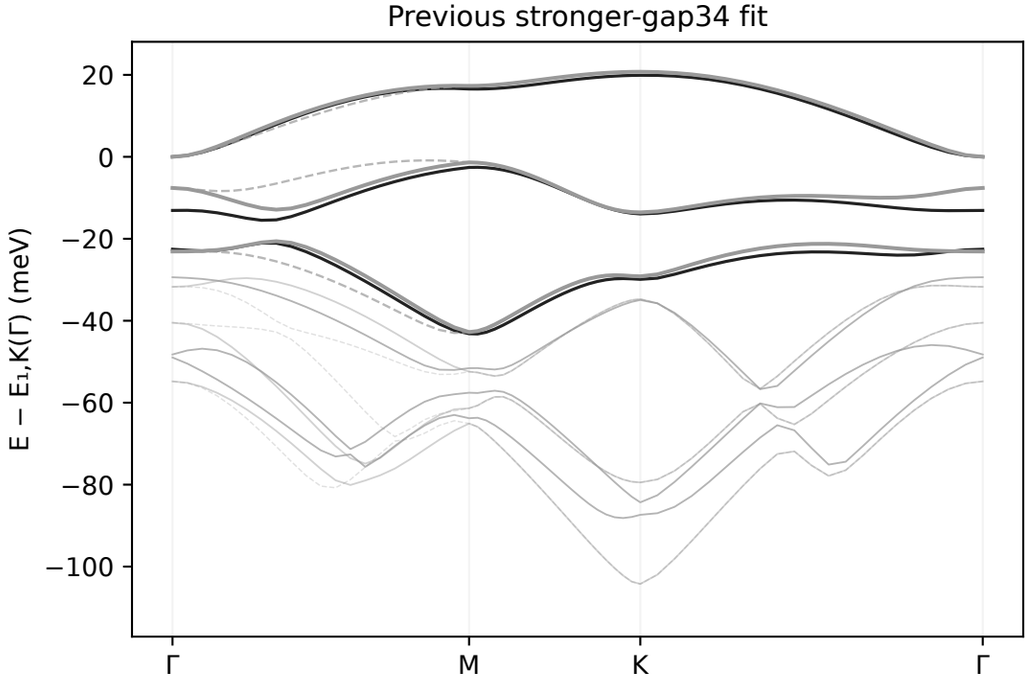
Top3 are thick and fitted. Bands4–6 are thin context. Opposite-valley Heff is not available in these snapshots.

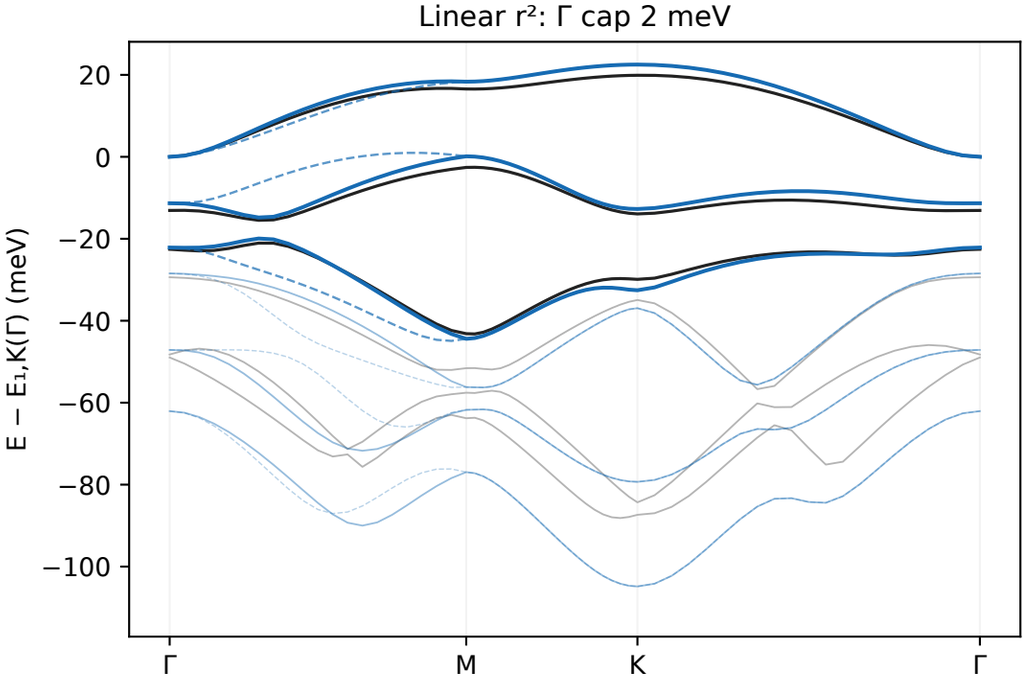
WSe₂ | 2.64591° | top3 repair with six-band context
All own models use pure quadratic kinetic energy and common mass/λ. Solid K; dashed K′; reference K only.

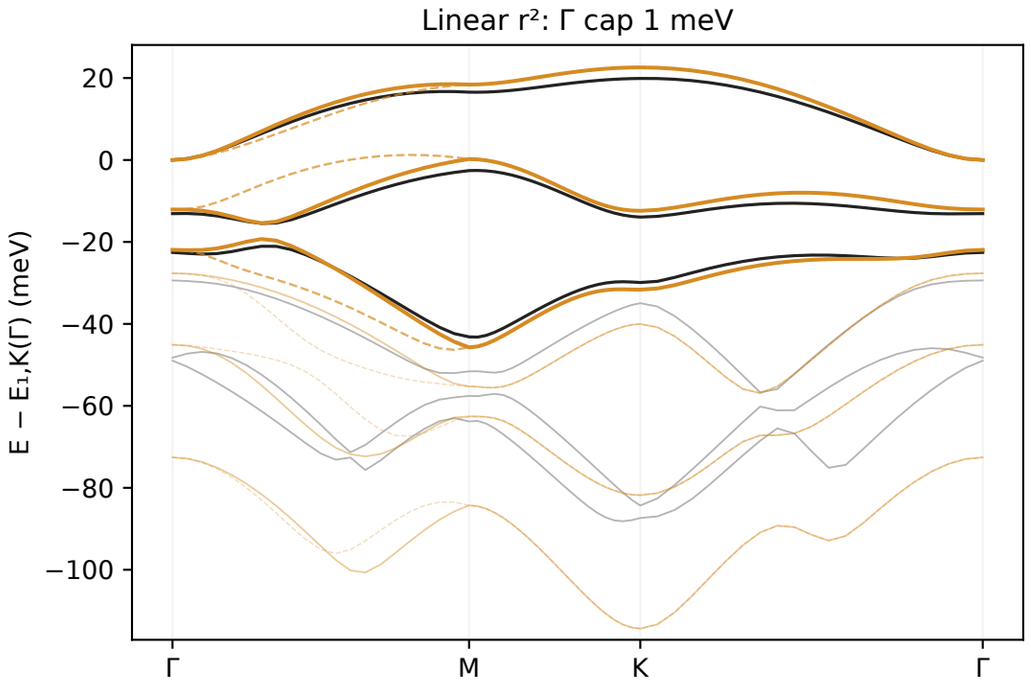


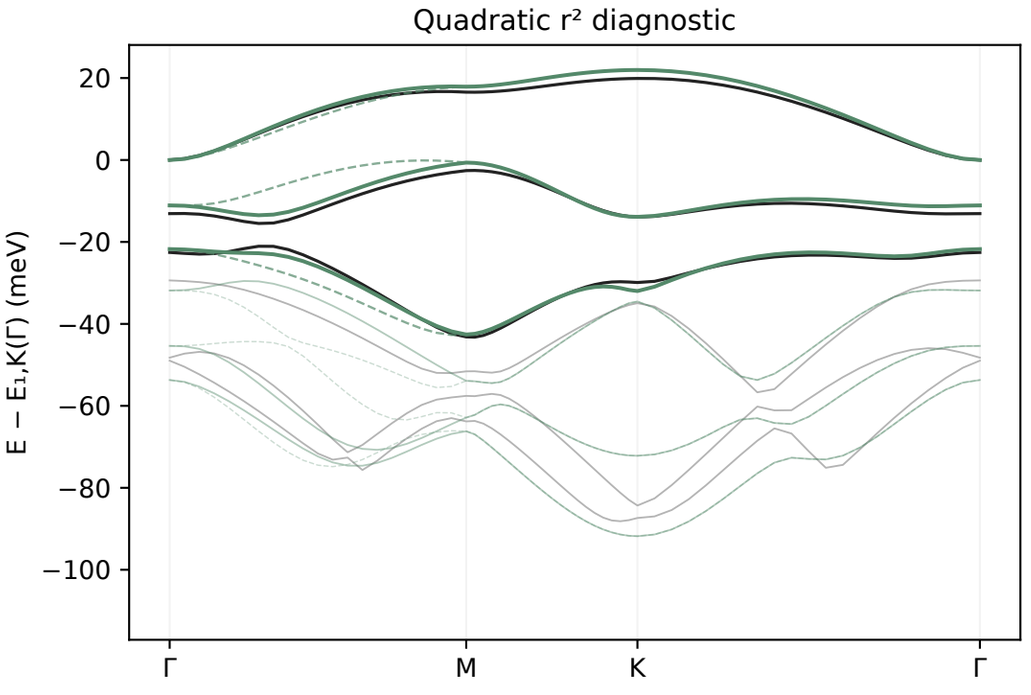
Top3 are thick and fitted. Bands4–6 are thin context. Opposite-valley Heff is not available in these snapshots.

WSe₂ | 2.87589° | top3 repair with six-band context
All own models use pure quadratic kinetic energy and common mass/λ. Solid K; dashed K′; reference K only.



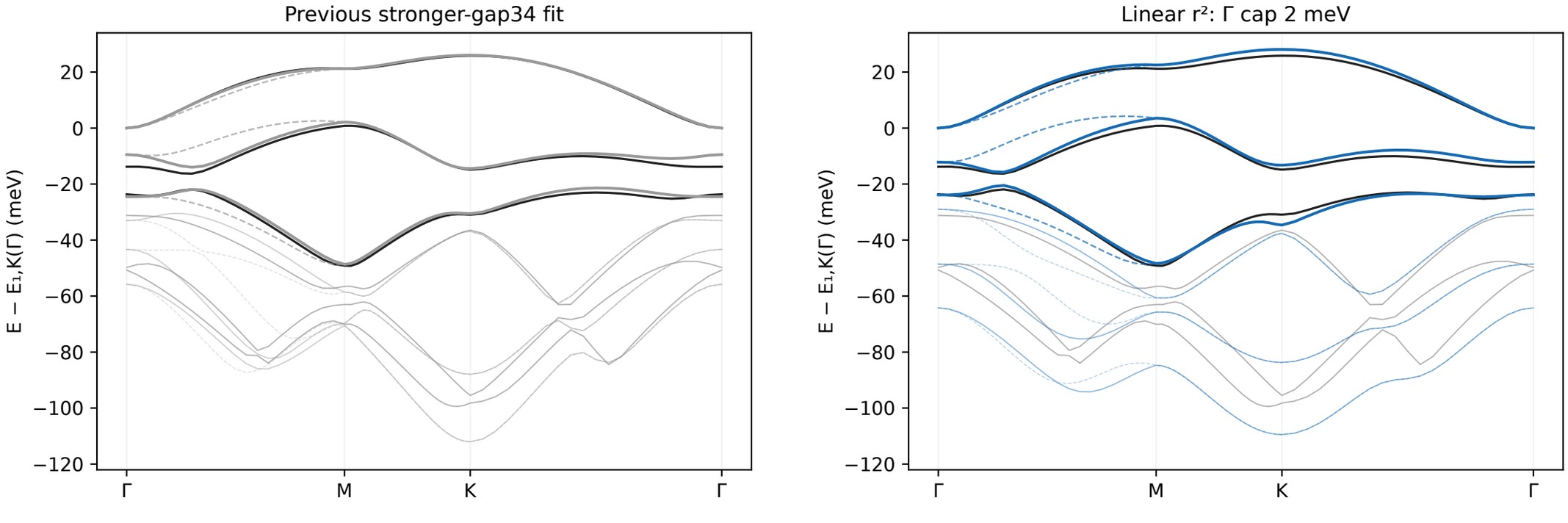






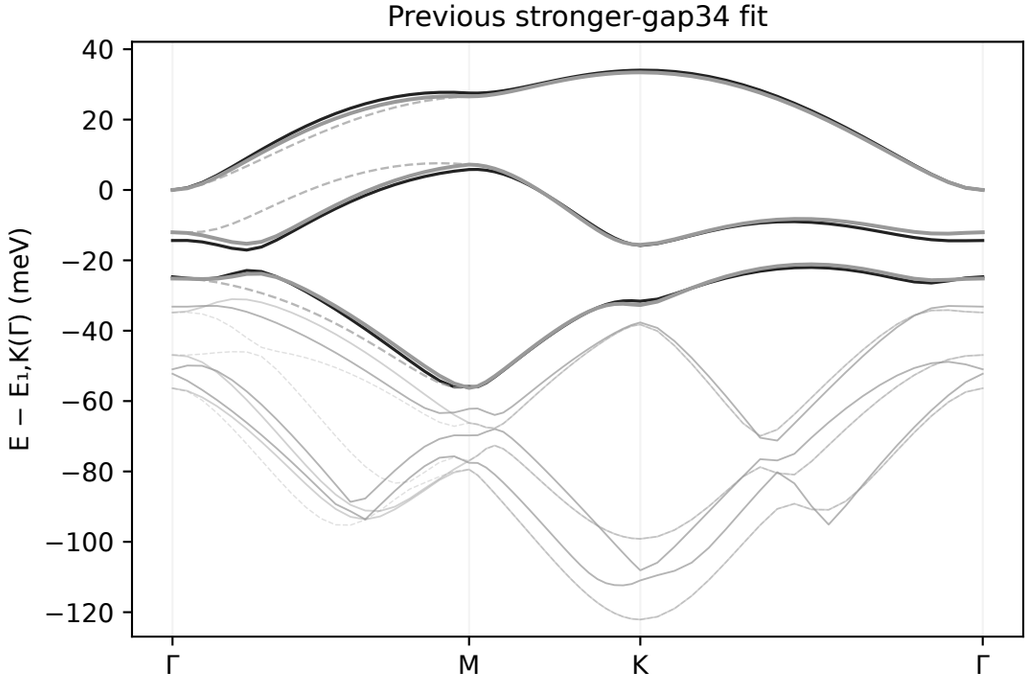
Top3 are thick and fitted. Bands4–6 are thin context. Opposite-valley Heff is not available in these snapshots.

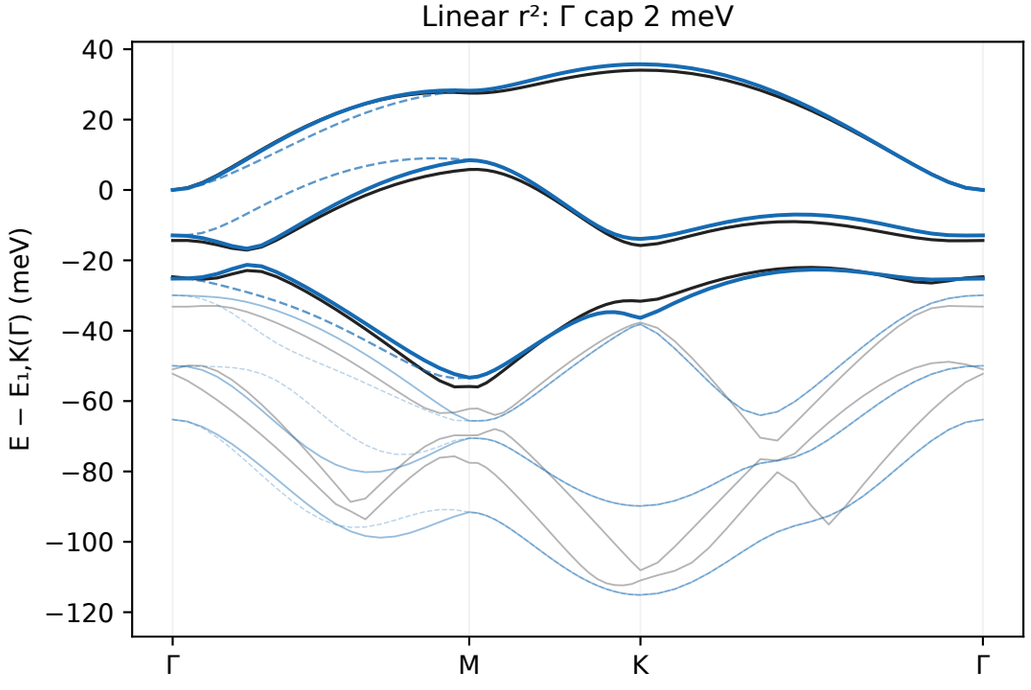
WSe₂ | 3.14966° | top3 repair with six-band context
All own models use pure quadratic kinetic energy and common mass/λ. Solid K; dashed K′; reference K only.

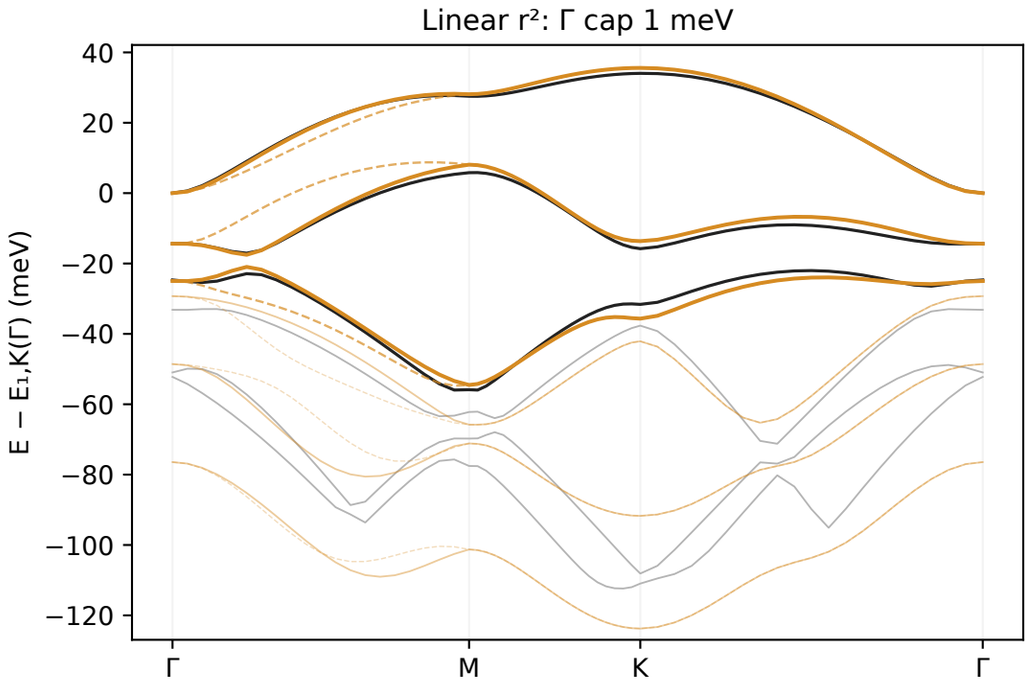


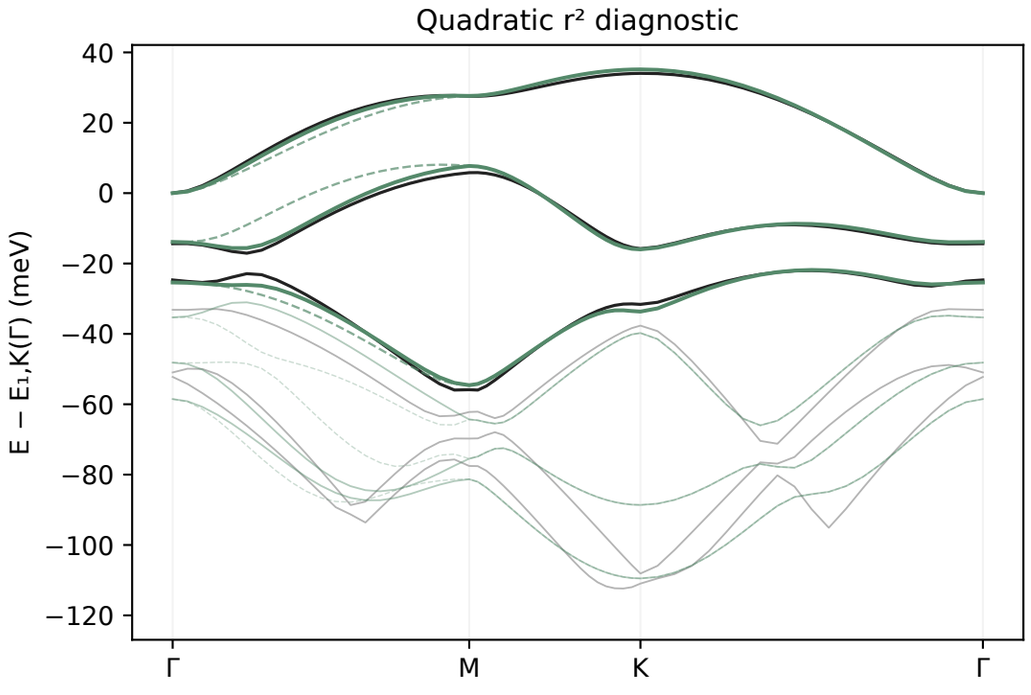
Top3 are thick and fitted. Bands4–6 are thin context. Opposite-valley Heff is not available in these snapshots.

WSe₂ | 3.48101° | top3 repair with six-band context
All own models use pure quadratic kinetic energy and common mass/λ. Solid K; dashed K′; reference K only.



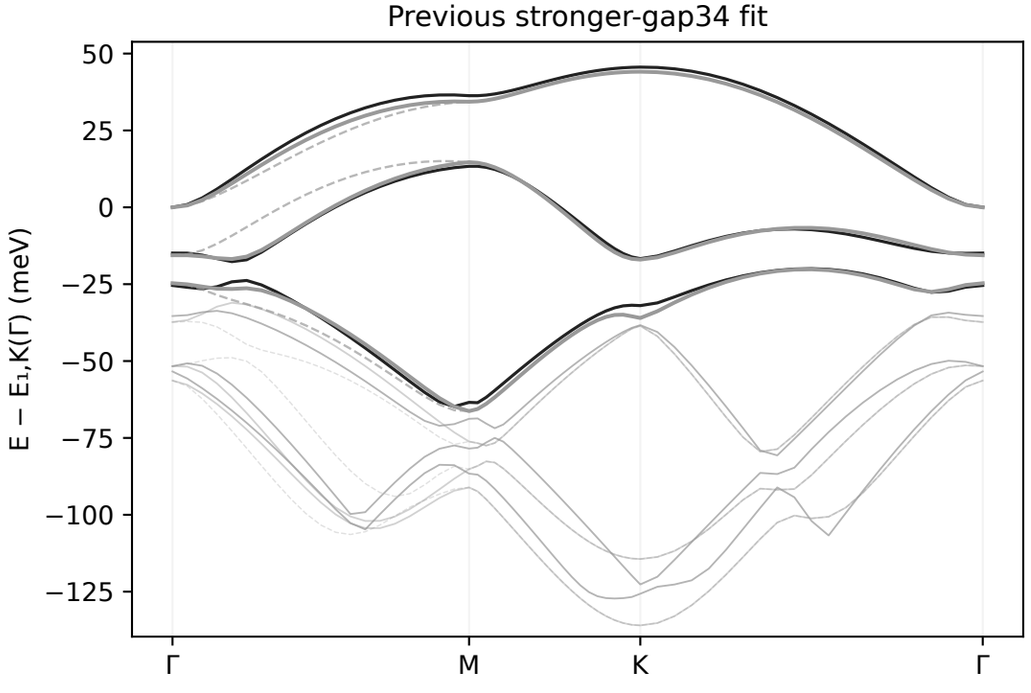


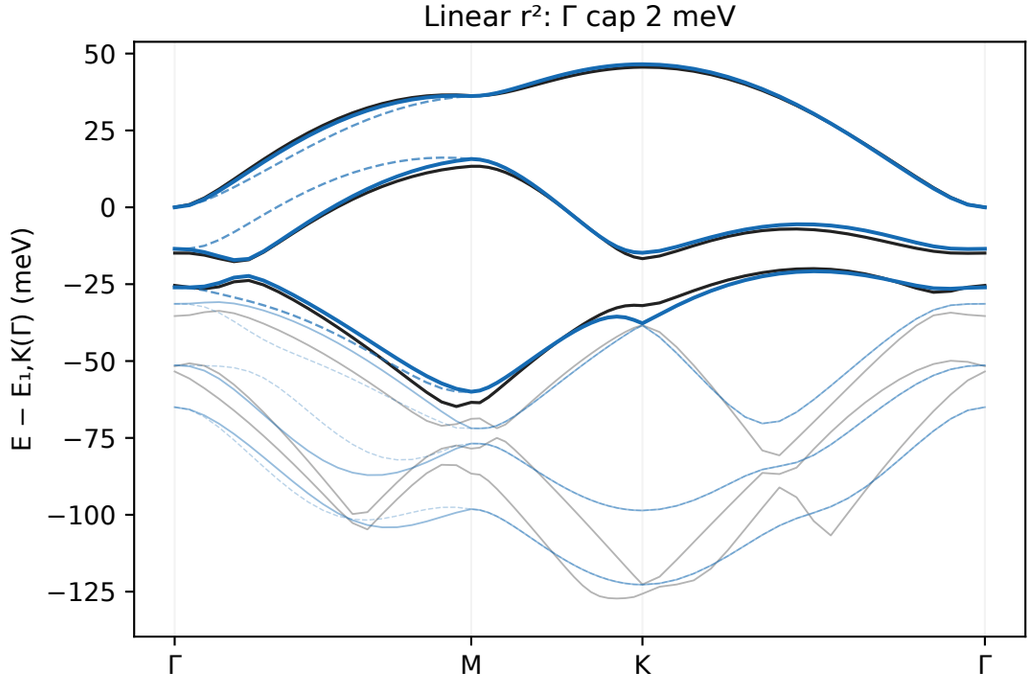


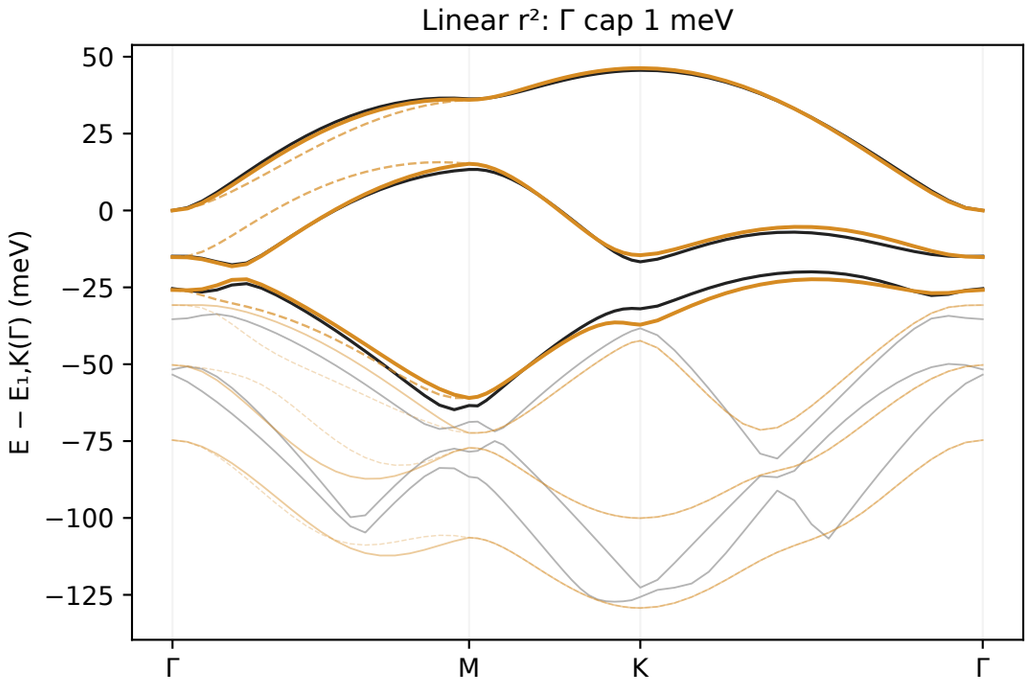


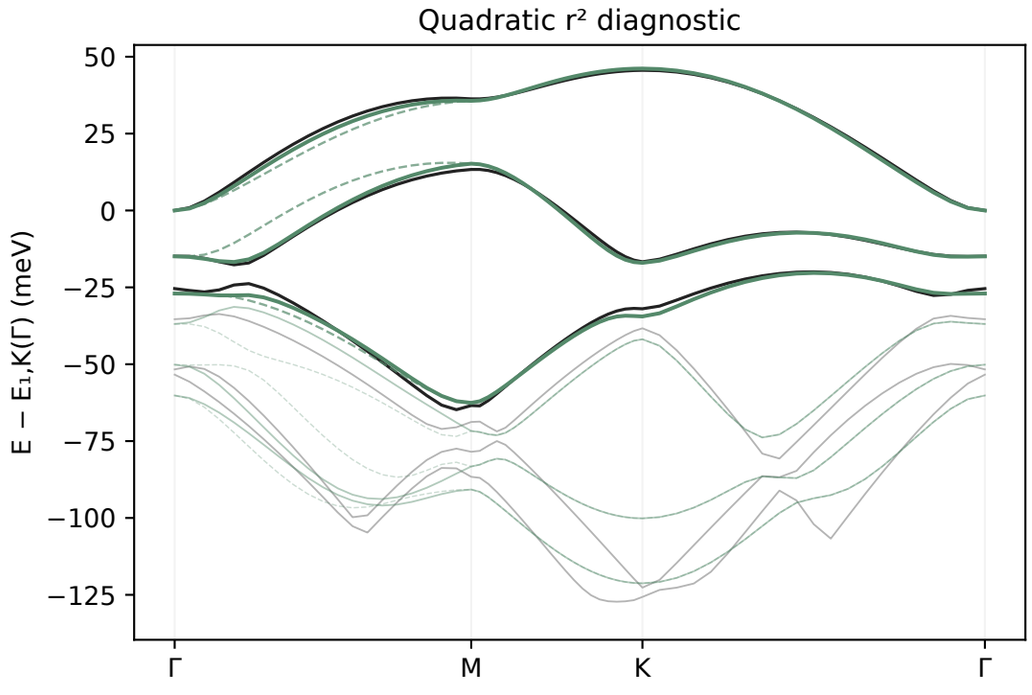
Top3 are thick and fitted. Bands4–6 are thin context. Opposite-valley Heff is not available in these snapshots.

WSe₂ | 3.89024° | top3 repair with six-band context
All own models use pure quadratic kinetic energy and common mass/λ. Solid K; dashed K′; reference K only.



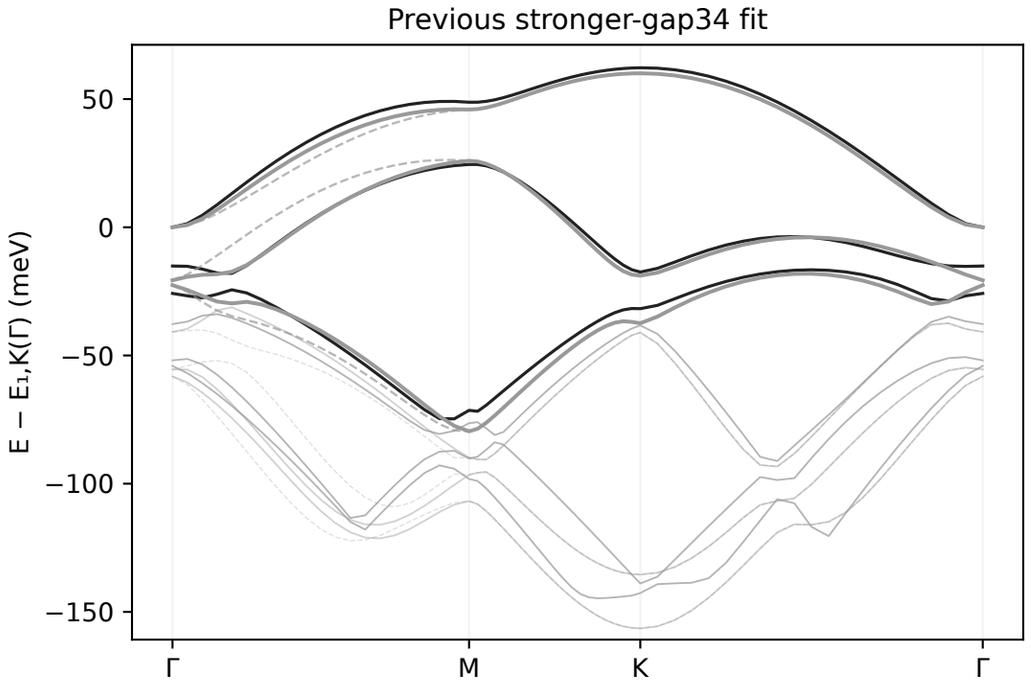


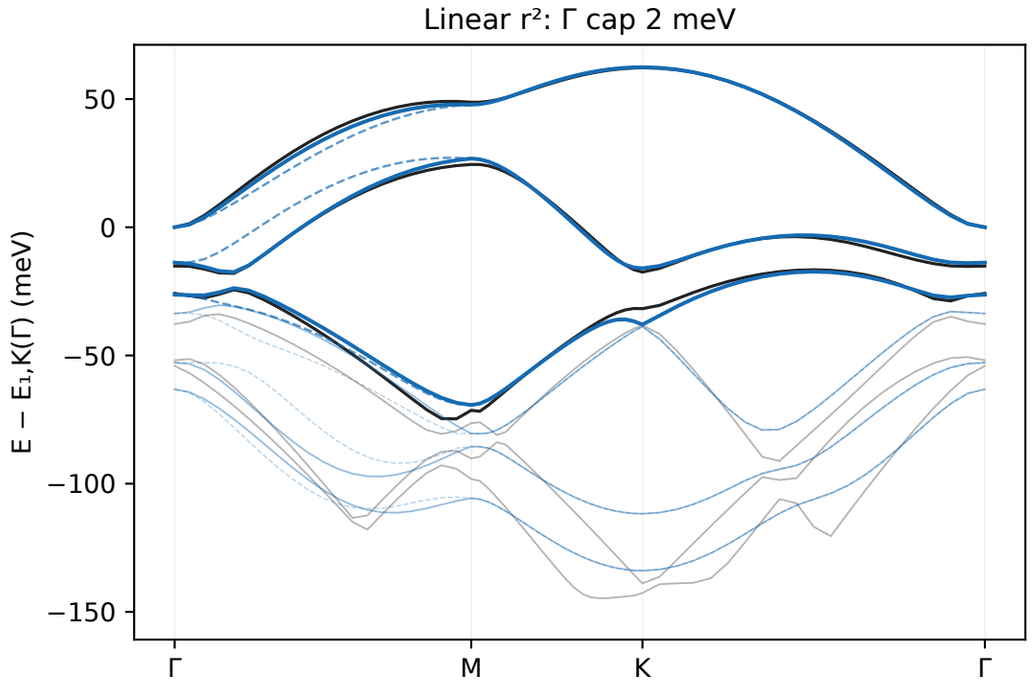


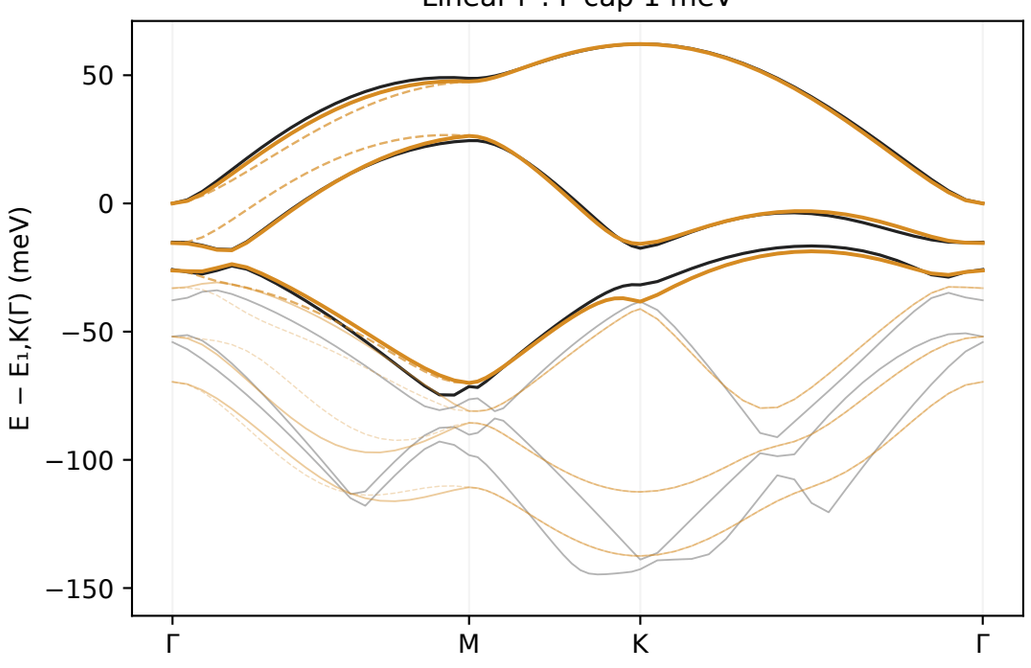


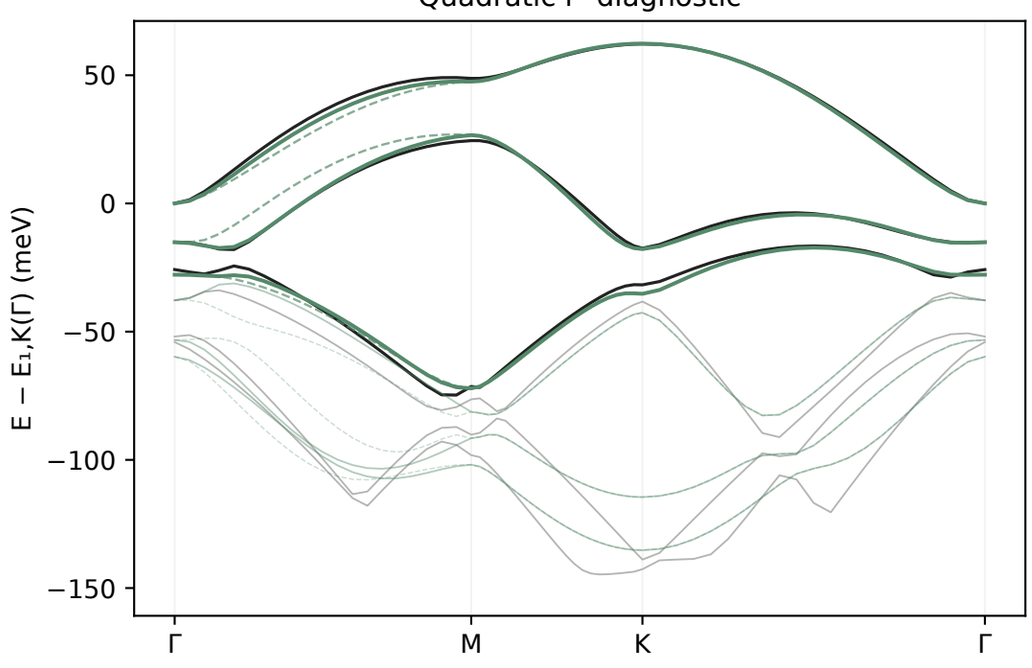
Top3 are thick and fitted. Bands4–6 are thin context. Opposite-valley Heff is not available in these snapshots.

WSe₂ | 4.40846° | top3 repair with six-band context
All own models use pure quadratic kinetic energy and common mass/λ. Solid K; dashed K′; reference K only.



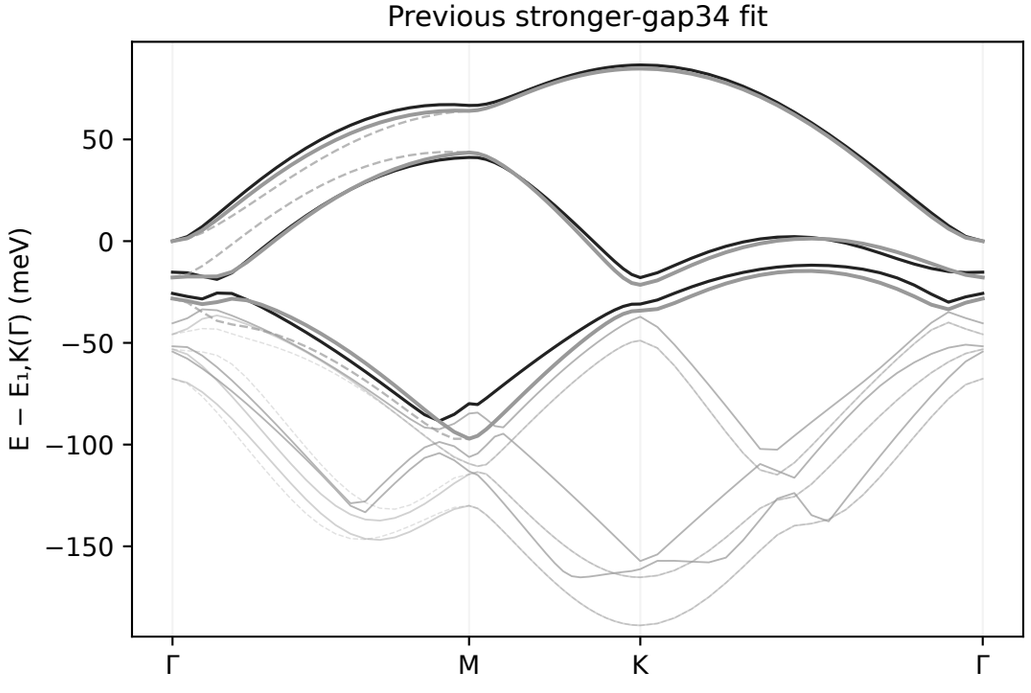


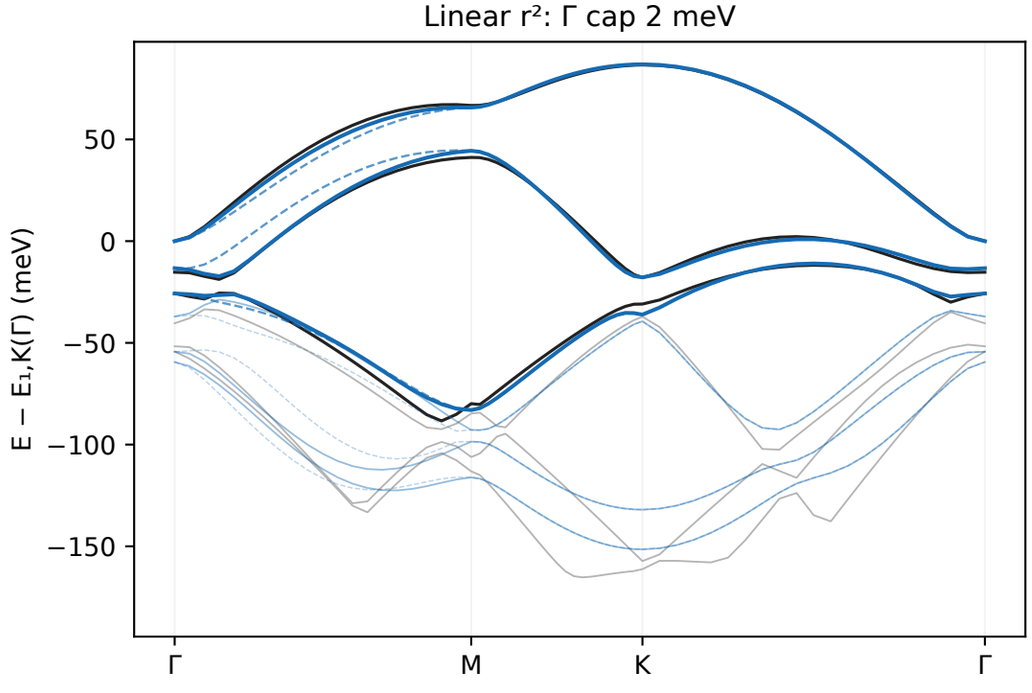


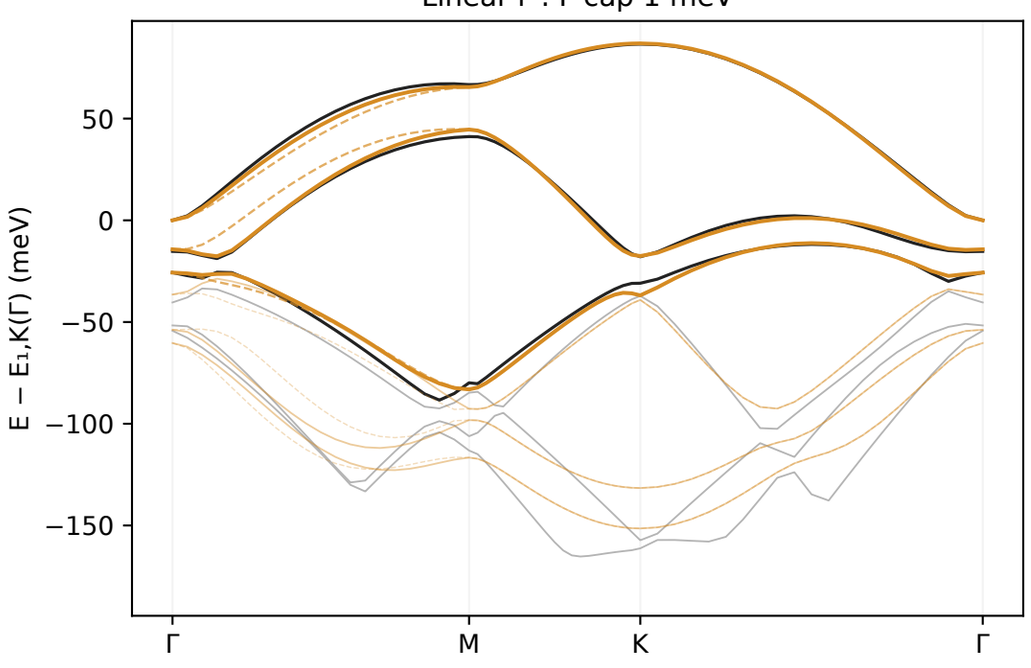


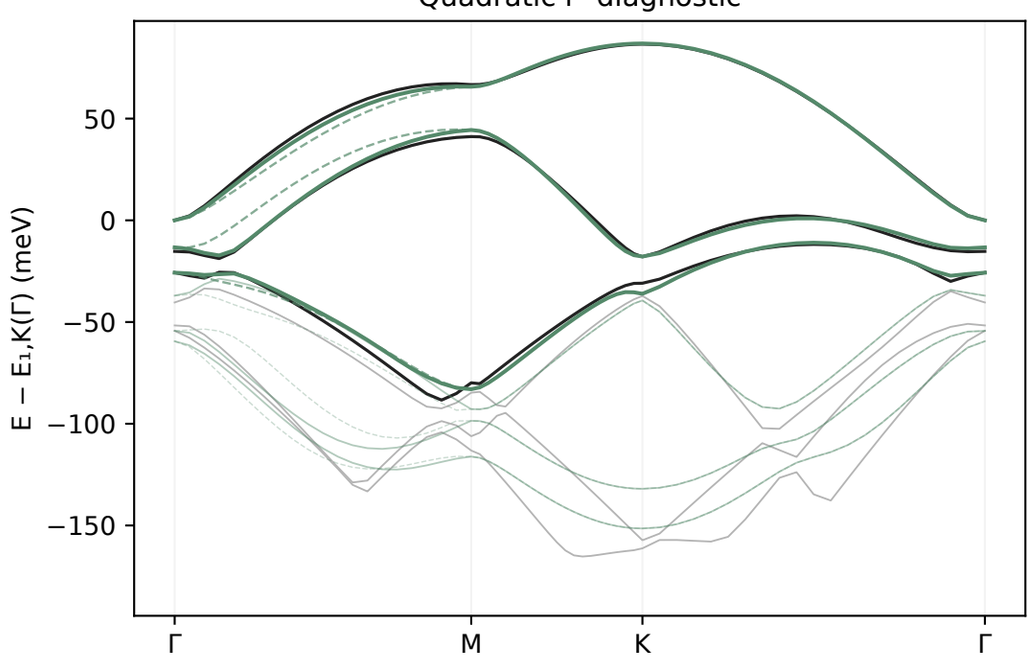
Top3 are thick and fitted. Bands4–6 are thin context. Opposite-valley Heff is not available in these snapshots.

WSe₂ | 5.08585° | top3 repair with six-band context
All own models use pure quadratic kinetic energy and common mass/λ. Solid K; dashed K′; reference K only.









Top3 are thick and fitted. Bands4–6 are thin context. Opposite-valley Heff is not available in these snapshots.